

Flag 1 – Install Flutter SDK

Objective: Install the Flutter SDK.

Verify:

```
flutter --version
```

Screenshot:

- The results of the command

```
Administrator: C:\windows\system32\cmd.exe - "C:\Users\Administrator\Desktop\src\flutter\bin\flutter.bat" --version - flutter --version

C:\Users\Administrator>flutter --version
Flutter 3.44.8 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 058e0af2c2 (7 days ago) • 2026-07-23 10:56:21 -0700
Engine • hash 13ffd72b2f9a5ca4db2a74ea52d5353ec2e8f939 (revision 0cd610717b) (6 days ago) • 2026-07-23 16:11:34.000Z
Tools • Dart 3.12.2 • DevTools 2.57.0

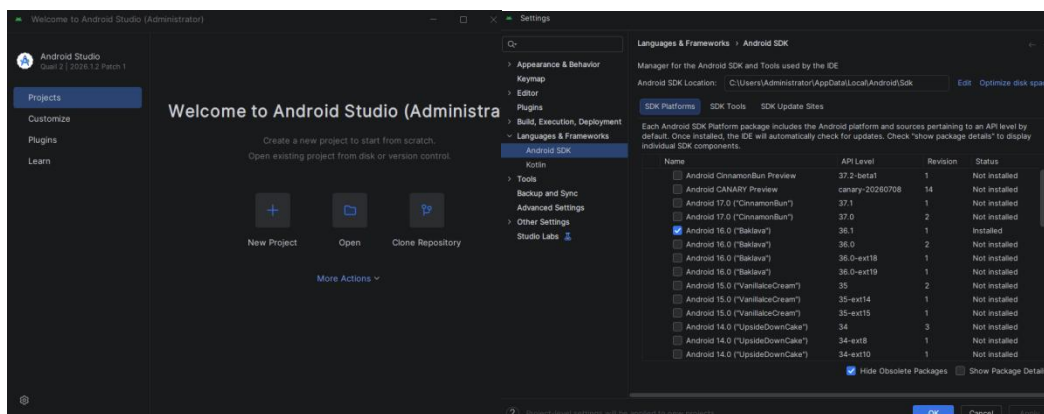
C:\Users\Administrator>Artajo
```

Flag 2 – Install Android Studio

Objective: Install Android Studio.wa

Screenshot:

- Android Studio installed
- SDK Manager opened (or Welcome screen)

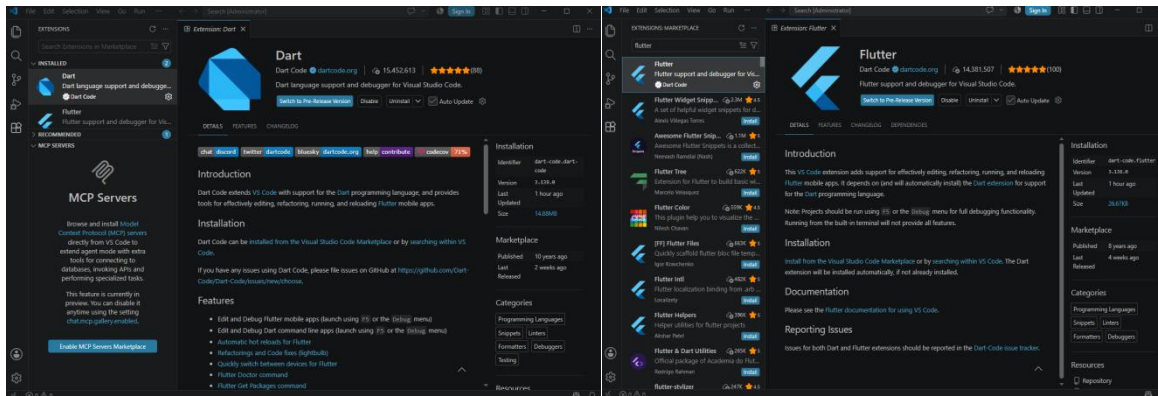


🚩 Flag 3 – Install VS Code & Flutter Extension

Objective: Install VS Code and the Flutter (which also installs Dart) extension.

Screenshot:

- Flutter extension installed
- Dart extension installed (or shown as installed)



🚩 Flag 4 – Run Flutter Doctor

Objective: Check whether everything is configured correctly.

Run:

`flutter doctor`

Screenshot:

- flutter doctor results

```
C:\Users\Administrator>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.8, on Microsoft Windows [Version 10.0.19045.7291], locale en-US)
[✓] Windows Version (10 Pro 64-bit, 22H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[✓] Connected device (2 available)
[✓] Network resources

! Doctor found issues in 1 category.

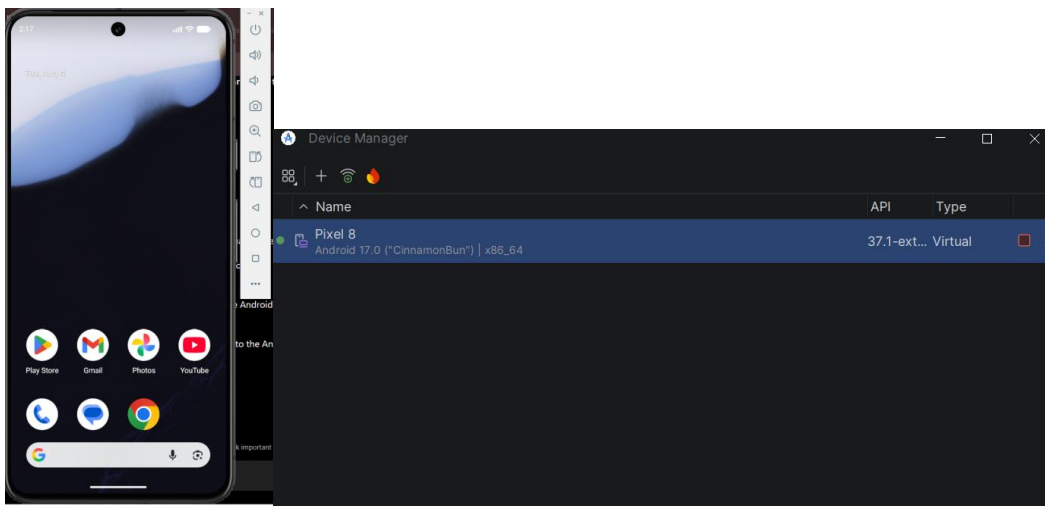
C:\Users\Administrator>
```

🚩 Flag 5 – Build Your Virtual Device

Objective: Create your first Android Emulator (AVD).

Screenshot:

- Device Manager
- Your newly created virtual device
- Emulator successfully booted



🚩 Flag 6 – Forge Your Project

Objective: Create your first Flutter project.

Run:

```
flutter create my_first_flutter_app
```

Screenshot:

- Successful creation of project

```

Microsoft Windows [Version 10.0.19045.7291]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>flutter create my_first_flutter_app
Creating project my_first_flutter_app...
Resolving dependencies in 'my_first_flutter_app'...
Downloading packages...
Got dependencies in 'my_first_flutter_app'.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd my_first_flutter_app
$ flutter run

Your application code is in my_first_flutter_app\lib\main.dart.

C:\Users\Administrator>

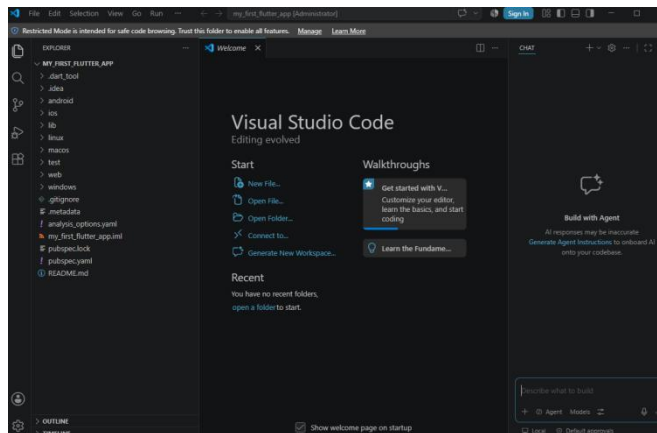
```

🚩 Flag 7 – Unlock the Workspace

Objective: Open the project in VS Code.

Screenshot:

- Your project opened in VS Code



🚩 Flag 8 – Launch

Objective: Run your Flutter application.

Run:

flutter run

Screenshot:

- Terminal
- Emulator running
- Default Flutter app

```
C:\Users\Ninja Turtle\Documents\my_first_flutter_app\lib>flutter run
Changing current working directory to: C:\Users\Ninja Turtle\Documents\my_first_flutter_app
Launching lib\main.dart on sdk gphone16k x86 64 in debug mode...
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by net.rubygrapefruit.platform.internal.NativeLibraryLoader in an unnamed module (file:/C:/Users/Ninja Turtle/.gradle/wrapper/dists/gradle-9.1.0-all/7wzd0jkjit61aq2p43wpjgij9/gradle-9.1.0/lib/gradle-native-platform-0.22-milestone-28.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for callers in this module
WARNING: Restricted methods will be blocked in a future release unless native access is enabled

Running Gradle task 'assembleDebug'...                               66.0s
✓ Built build\app\outputs\flutter-apk\app-debug.apk
Installing ..\build\app\outputs\flutter-apk\app-debug.apk...       4.9s
I/FlutterActivityAndFragmentDelegate(11192): If you are attempting to set --enable-dart-profiling via Intent extras to launch a Flutter component outside of using the Flutter CLI, note that support for setting engine flags on Android via Intent will soon be dropped; see https://github.com/flutter/flutter/issues/180686 for more information on this breaking change. To migrate, set --enable-dart-profiling or any other flags specified via Intent extras on the command line instead or see https://github.com/flutter/flutter/blob/main/docs/engine/Flutter-Android-Engine-Flags.md for alternative methods.
D/FlutterJNI(11192): Beginning load of flutter...
D/FlutterJNI(11192): flutter (null) was loaded normally!
I/flutter (11192): [IMPORTANT:flutter/shell/platform/android/android_context_gl_impeller.cc(104)] Using the Impeller rendering backend (OpenGL ES).
D/FlutterRenderer(11192): Width is zero. 0,0
Syncing files to device sdk gphone16k x86 64...                   112ms

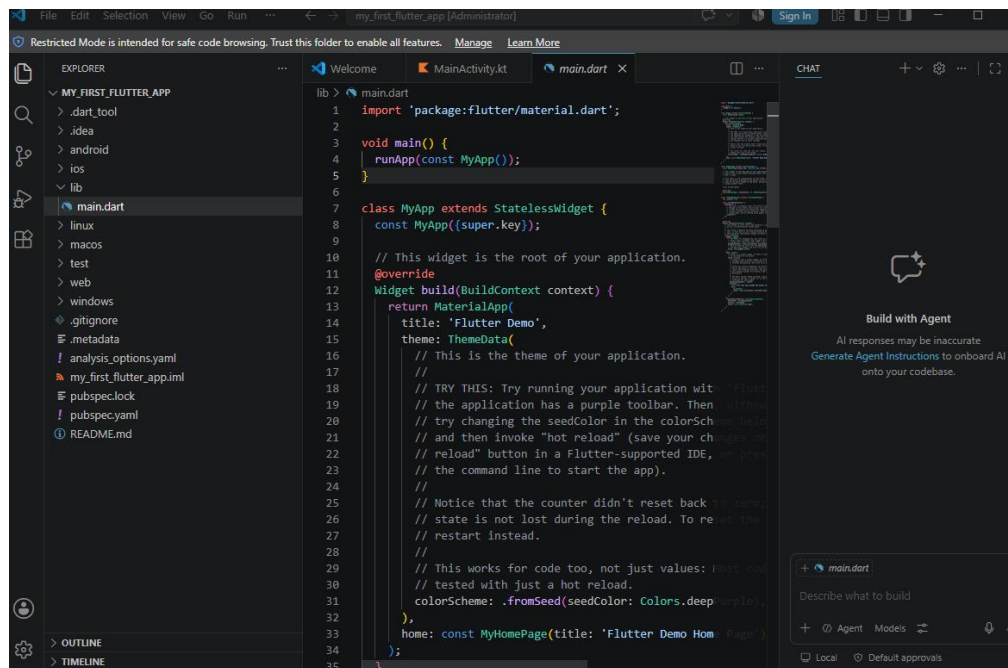
Flutter run key commands.
r Hot reload.
R Hot restart.
```

🚩 Flag 9 – Find the Entry Point

Figure out where the default Flutter page comes from.

Screenshot:

- The location of the file where the default page is from (the one showing on the emulator)



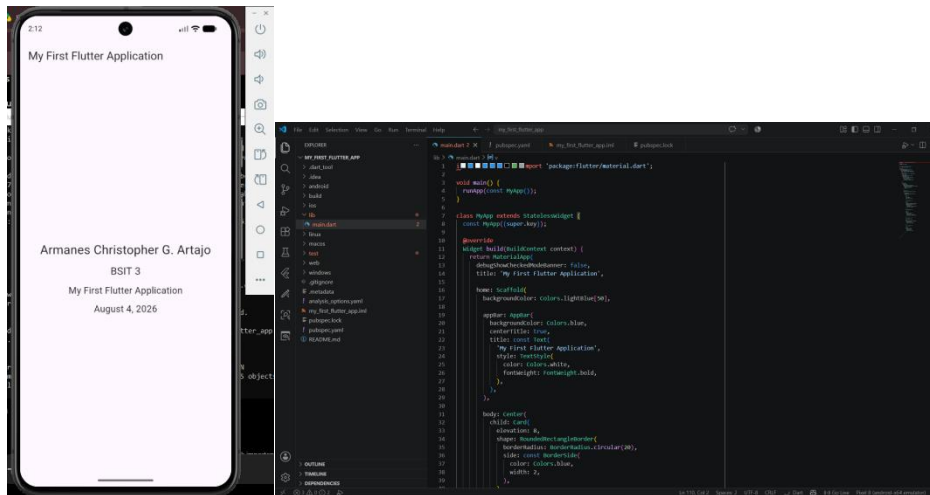
🚩 Flag 10 – Claim Your App

Replace the default screen with:

- Full Name
- Course & Section
- "My First Flutter Application"
- Today's Date

Screenshot:

- Your updated content of the code
- The updated UI



Flag 11 – Personalize It

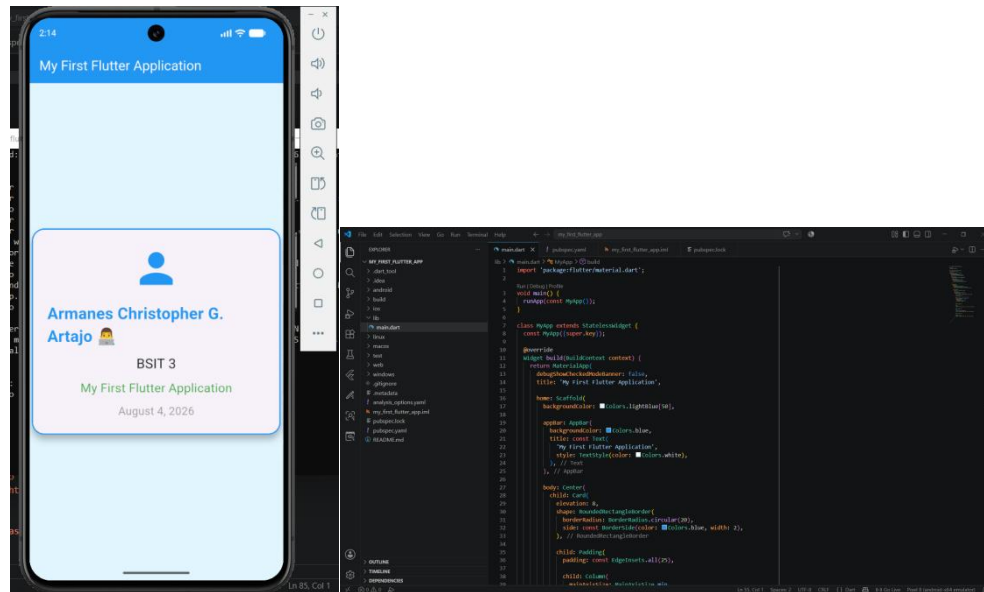
Research and implement at least **three** visual changes.

Examples:

- Change the background colour.
- Change the text colour.
- Centre the content.
- Increase font size.
- Add a border or rounded card.
- Change the font.
- Add an emoji or icon.

Screenshot:

- The code changed for personalizing
- The updated UI



🚩 Flag 12 – Add an Image

Research how to:

- Create an **assets** folder
- Register assets
- Display an image

Screenshot:

- The steps it took to add an image
- The updated UI

